Trustworthy AI for Open RAN — Analysis Report

Audience: AI research

Generated:
[cci_a1sd_binary] xgboost — accuracy=1.000
Personal research project — not affiliated with any employer.

Contents

- Executive summary
- Dataset & model
- Base performance
- Explainability
- Robustness
- Fairness
- Privacy
- O-RAN console snapshot
- Agent trust evaluation
- Methodology & validity limits

Executive summary

Methodological summary: each figure below comes from a single TrustPipeline run on the held-out test split (seed 42), optionally paired with a trace-level trustagent evaluation over 0 cases.

Key read-out

Metric	Value
Accuracy	0.9996
F1 (macro)	0.9996
Roc auc	0.9998
Pgd attack success rate	0.32
Membership inference auc	0.4946

Reported quantities are point estimates without confidence intervals. Seeds are fixed so reruns are reproducible, not so the numbers generalise across datasets.

Dataset & model

Dataset

Metric	Value
Dataset	cci_a1sd_binary
Task	binary
Samples	24000
Features	10
Classes	Normal, Dos
Group attrs	policy_type_id, load_label, operation, policy_name
Class counts	Normal=12000; Dos=12000

The dataset describes A1 traffic: Normal vs DoS.

Model configuration

Metric	Value
Model	xgboost
Task	binary
Params	n_estimators=200; max_depth=6; learning_rate=0.1; subsample=0.9; colsample_bytree=0.9; random_state=42
Elapsed sec	5.95

Feature list

Metric	Value
Feature names	real_interval, inter_arrival_mean, inter_arrival_std, inter_arrival_cv, response_time_mean, response_time_std …
Meta	source=CCI-A1SD; dos_variant=dos2; classes=normal, dos; per_class_cap=20000; imputed_readings=0; policy_names=20008=Traffic Steering; 20009=QoS Management; 20010=Resource Allocation; 20011=Slice SLA; 20012=Energy Saving; 20013=Load Balancing

Base performance

Headline metrics on the held-out test split.

Metric	Value
Accuracy	0.9996
F1 (macro)	0.9996
Roc auc	0.9998

Figures are rendered from the same run the panels display.

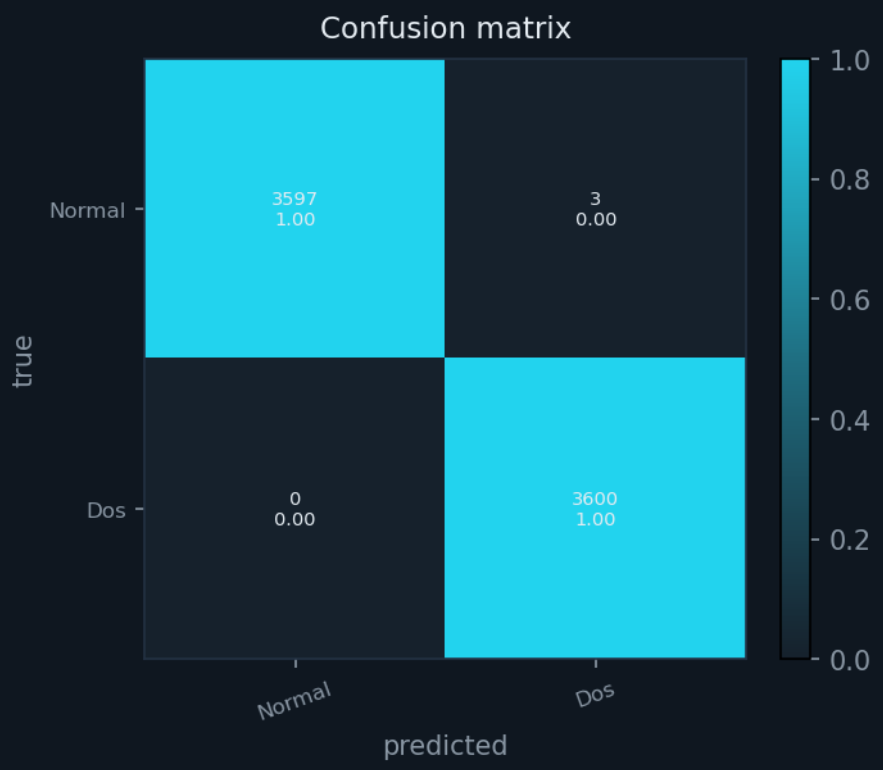


Class distribution

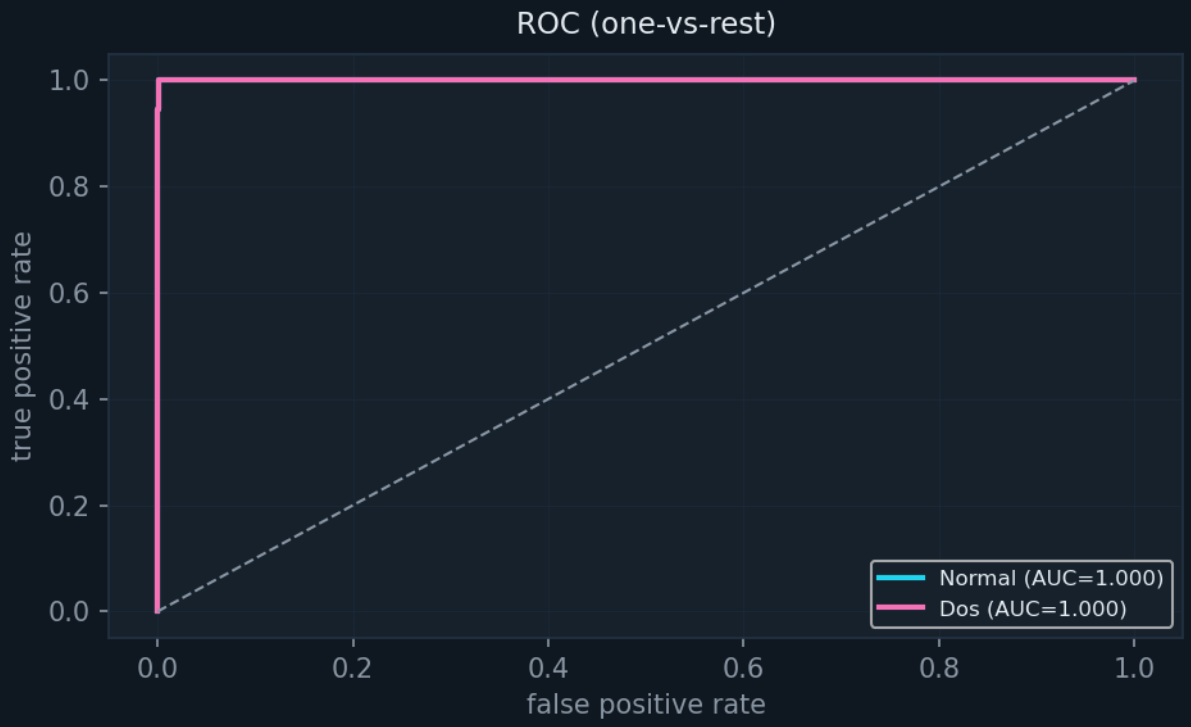
cci_a1sd — feature distributions



Feature histograms



Confusion matrix



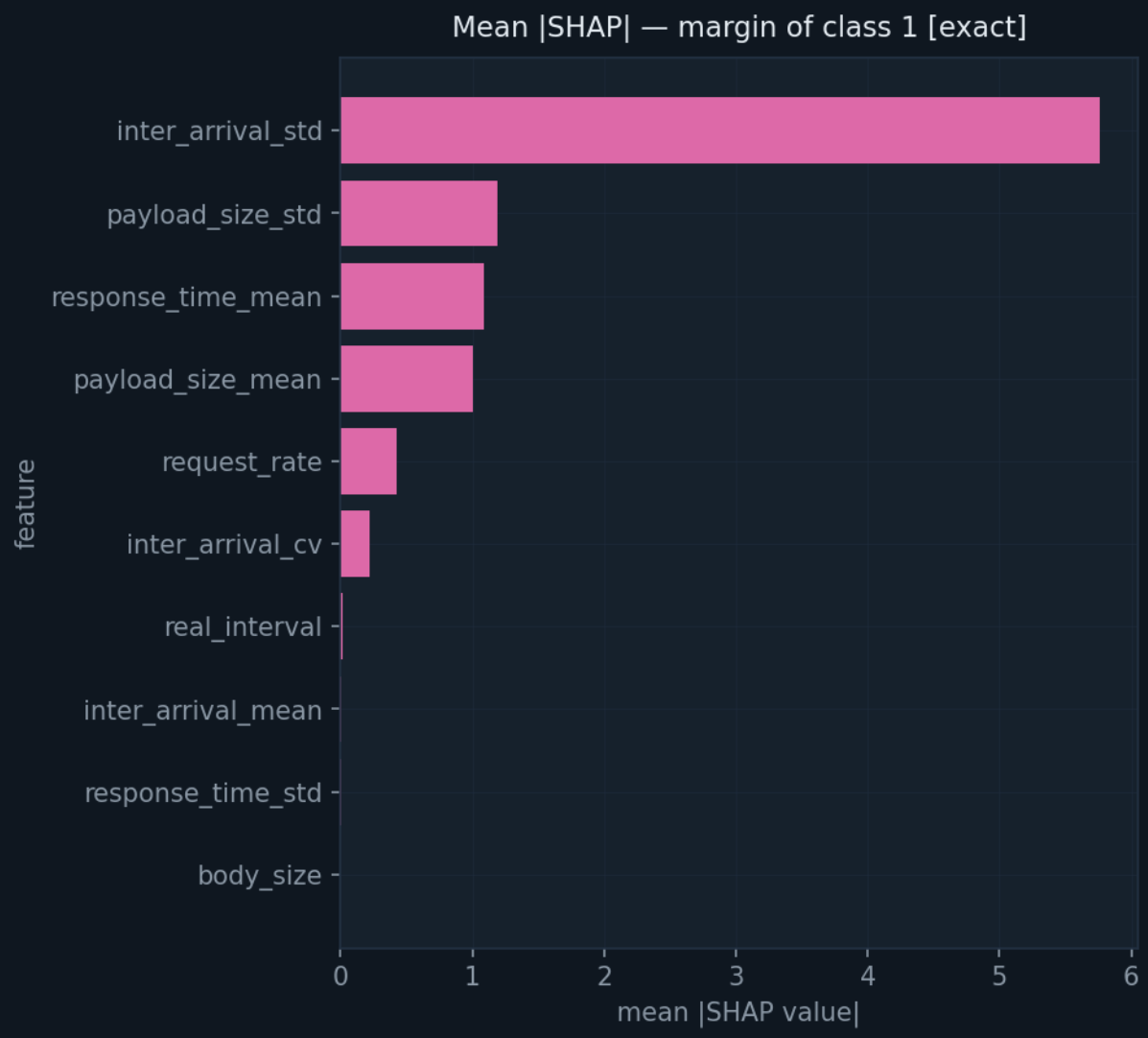
ROC

Explainability

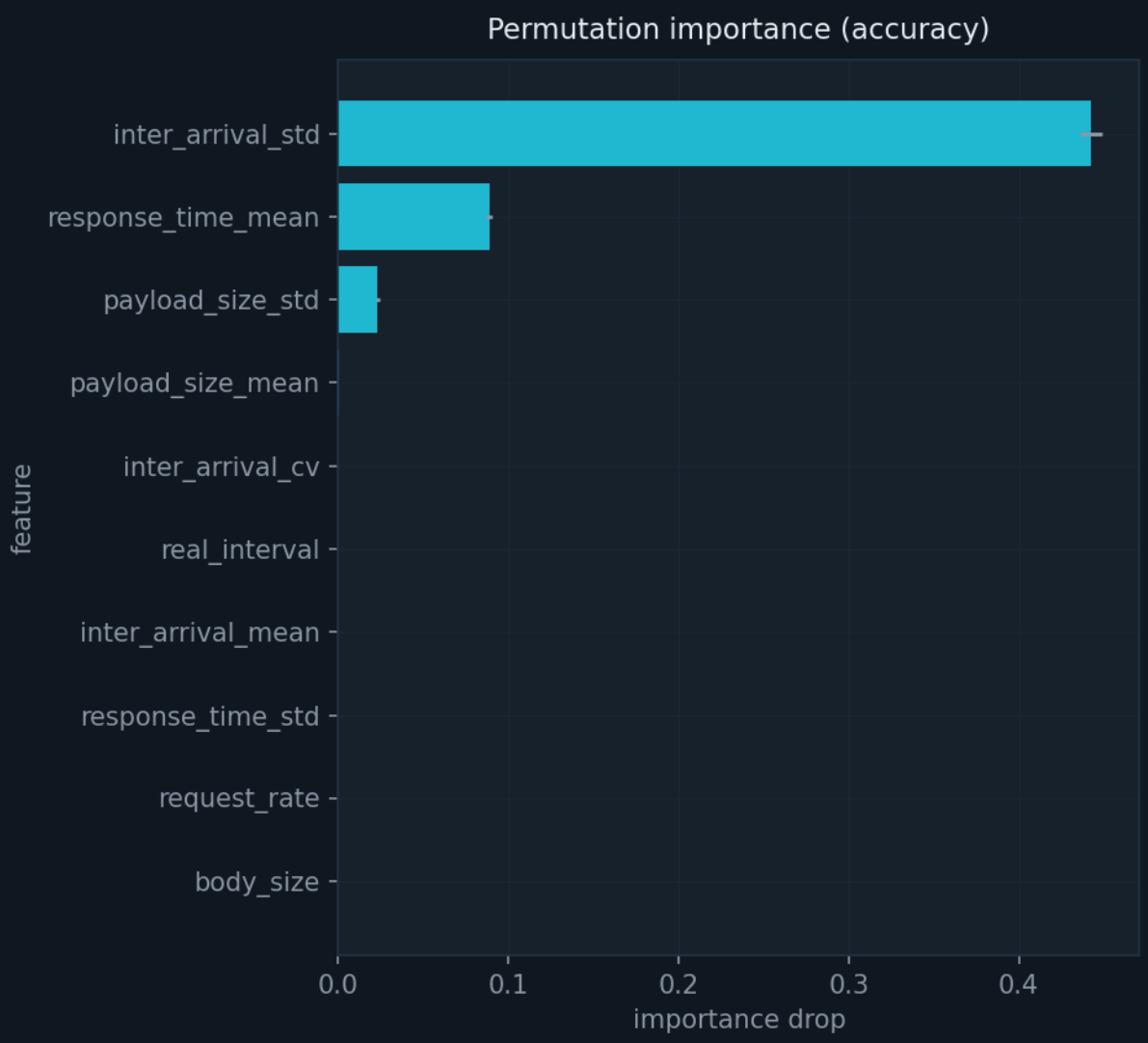
Global importance (permutation), KernelSHAP-style attributions, a LIME-style local surrogate and PDP/ICE curves, all computed on the test split.

Metric	Value
Top features	inter_arrival_std, response_time_mean, payload_size_std, payload_size_mean, inter_arrival_cv

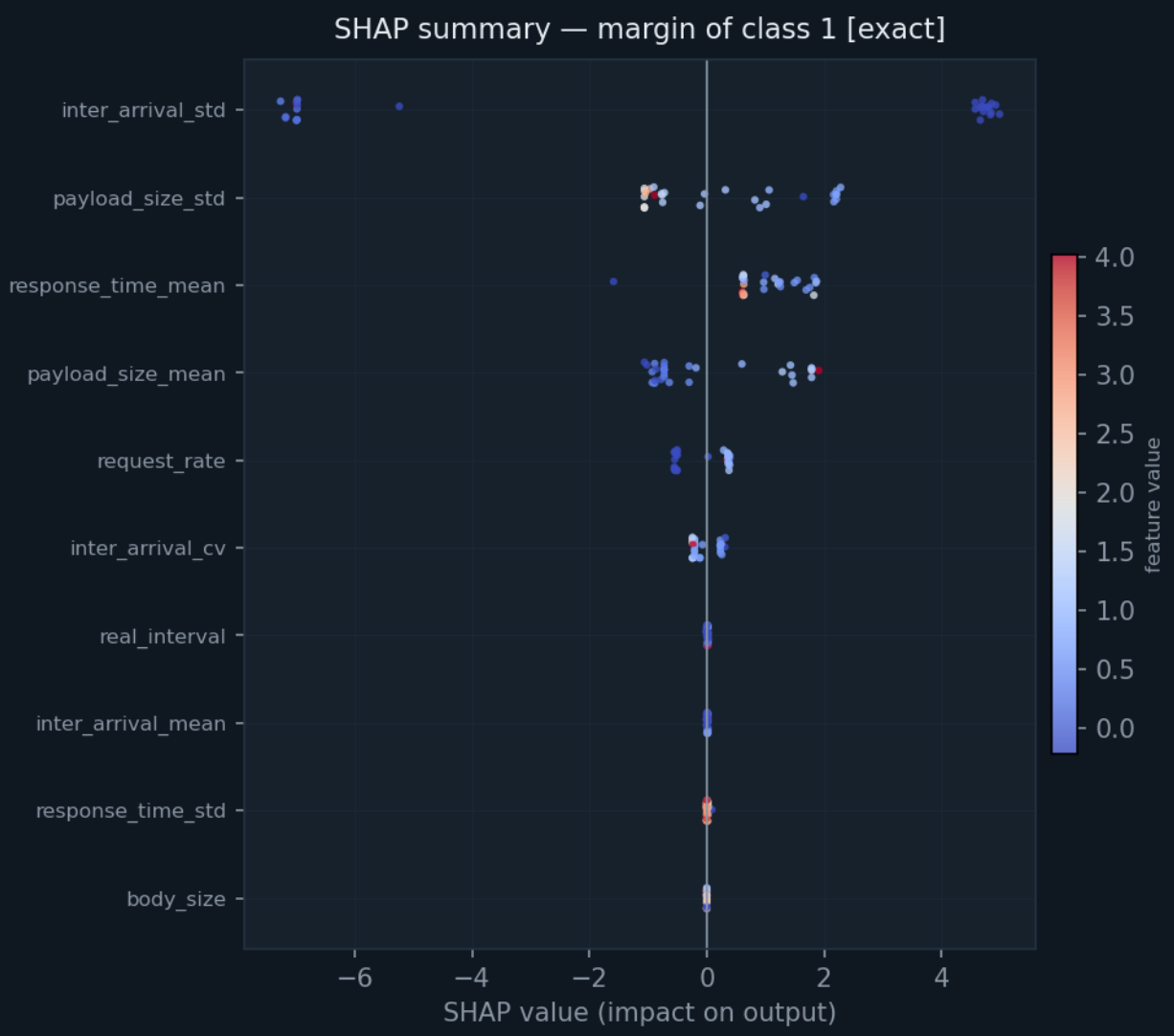
Metric	Value
Target	margin of class 1
Shap top	inter_arrival_std, payload_size_std, response_time_mean, payload_size_mean, request_rate
Method	permutation + exact TreeSHAP (native pred_contribs) + LIME-style local + PDP/ICE
Shap method	exact
Shap efficiency max error	6.284e-06



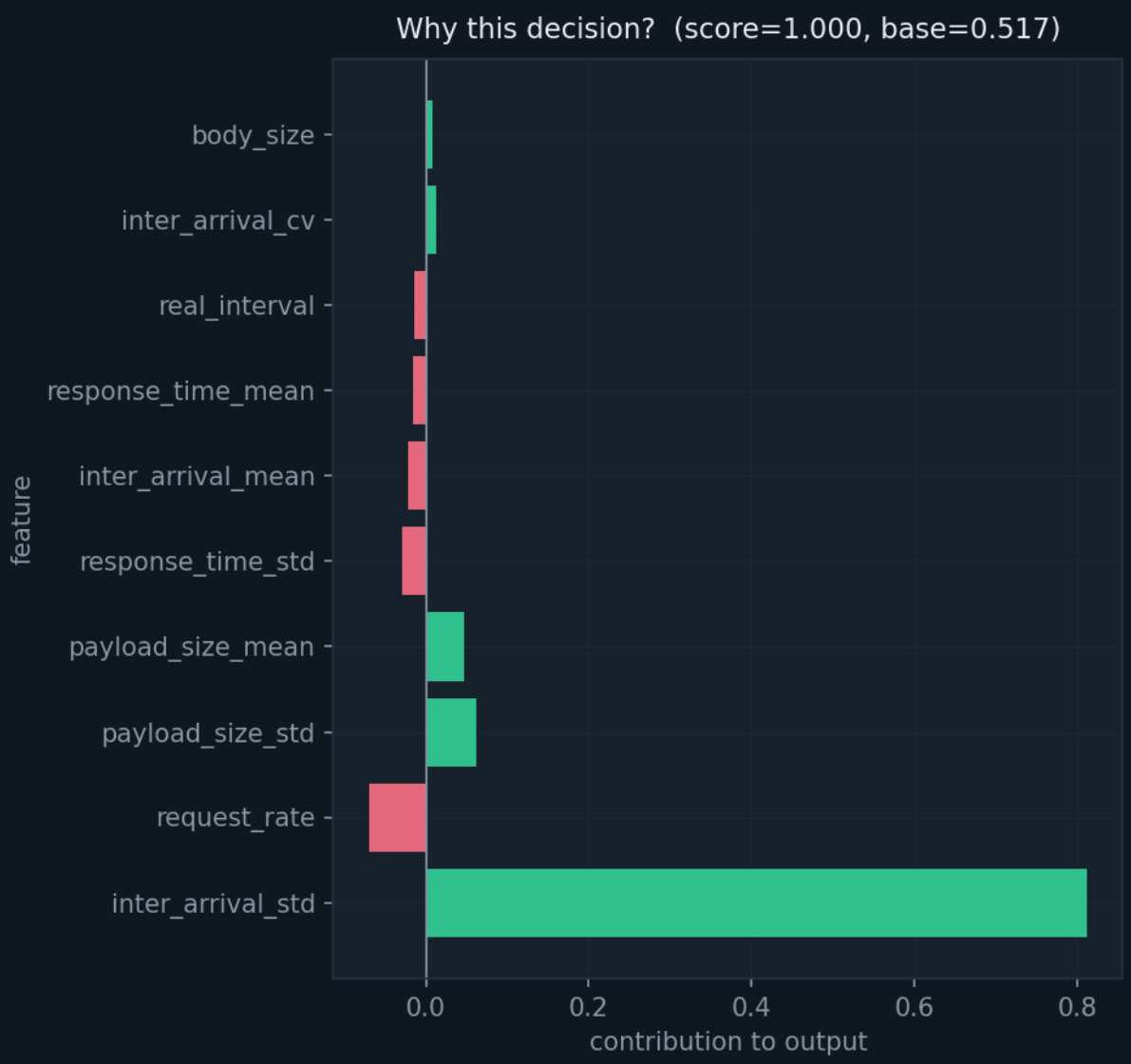
Shap bar



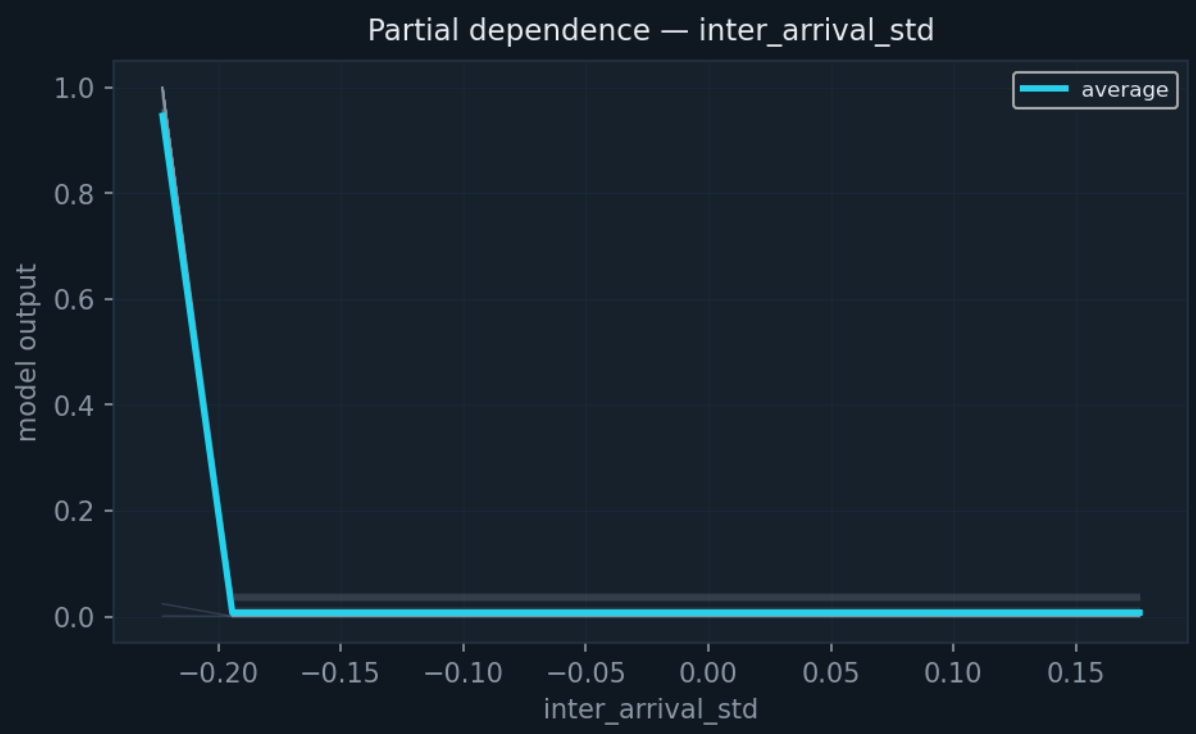
Permutation importance



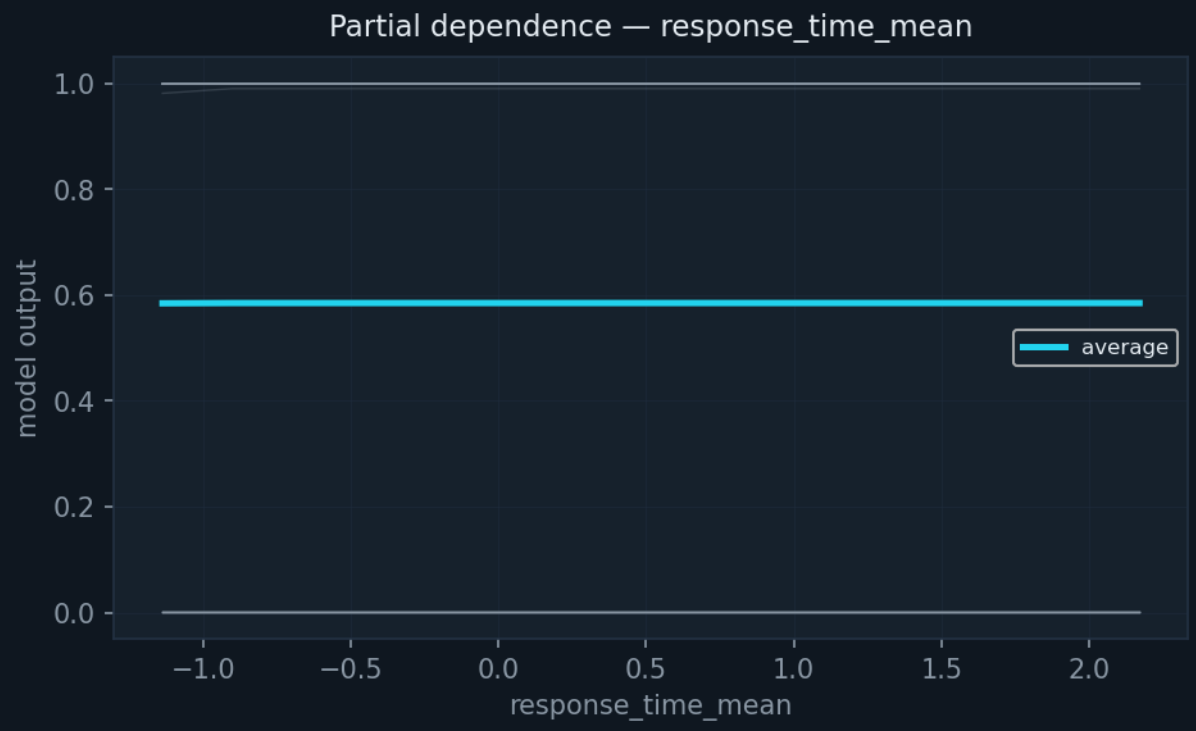
Shap summary



Local explanation



PDP · inter-arrival jitter



PDP · mean response time

Permutation importance

feature	permutation	permutation_std	native
inter_arrival_std	0.442	0.006529	0.5206
response_time_mean	0.08872	0.001627	0.1625
payload_size_std	0.02316	0.0012	0.0201
payload_size_mean	0.0004514	0.000248	0.007406
inter_arrival_cv	6.944e-05	6.944e-05	0.001617

feature	permutation	permutation_std	native
real_interval	0	0	0.00152
inter_arrival_mean	0	0	0.001576
response_time_std	0	0	0.001531
request_rate	0	0	0.2832
body_size	0	0	0

SHAP mean |value|

feature	mean_abs_shap
inter_arrival_std	5.761
payload_size_std	1.187
response_time_mean	1.087
payload_size_mean	1.001
request_rate	0.4253
inter_arrival_cv	0.2207
real_interval	0.01131
inter_arrival_mean	0.01047
response_time_std	0.009575
body_size	0

Local explanation

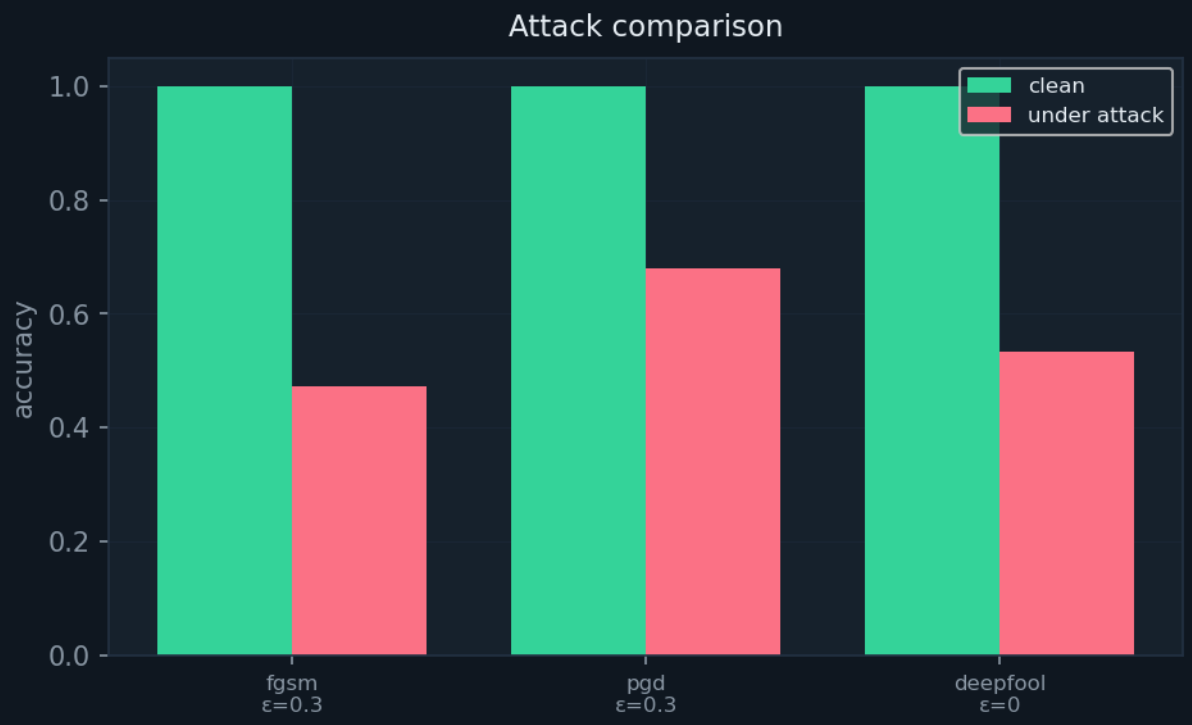
feature	contribution
inter_arrival_std	0.8122
request_rate	-0.07009
payload_size_std	0.06142
payload_size_mean	0.04718
response_time_std	-0.02905
inter_arrival_mean	-0.02265
response_time_mean	-0.01553
real_interval	-0.01399
inter_arrival_cv	0.01183
body_size	0.008597

Robustness

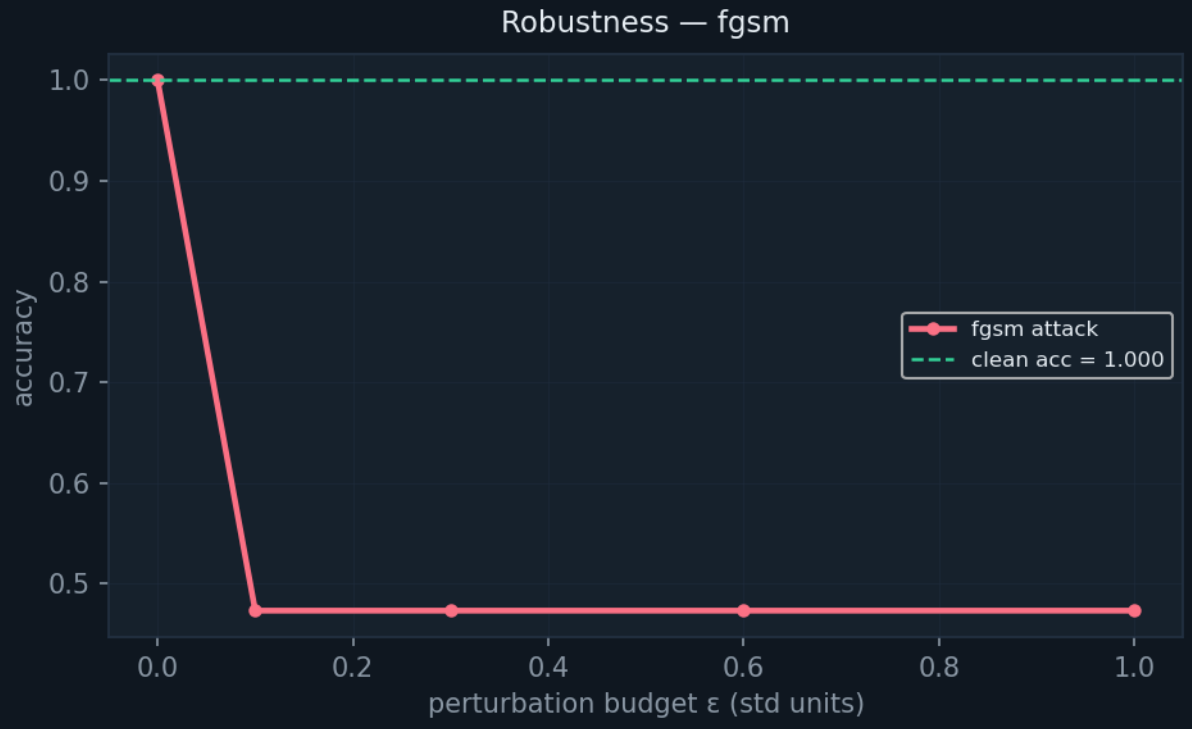
FGSM / PGD / DeepFool via surrogate gradients, robustness curves over an ε sweep, and adversarial training as a defence baseline.

Metric	Value
Clean accuracy	1
Fgsm robust acc	0.4733
Pgd robust acc	0.68
Deepfool robust acc	0.5333
Pgd attack success rate	0.32

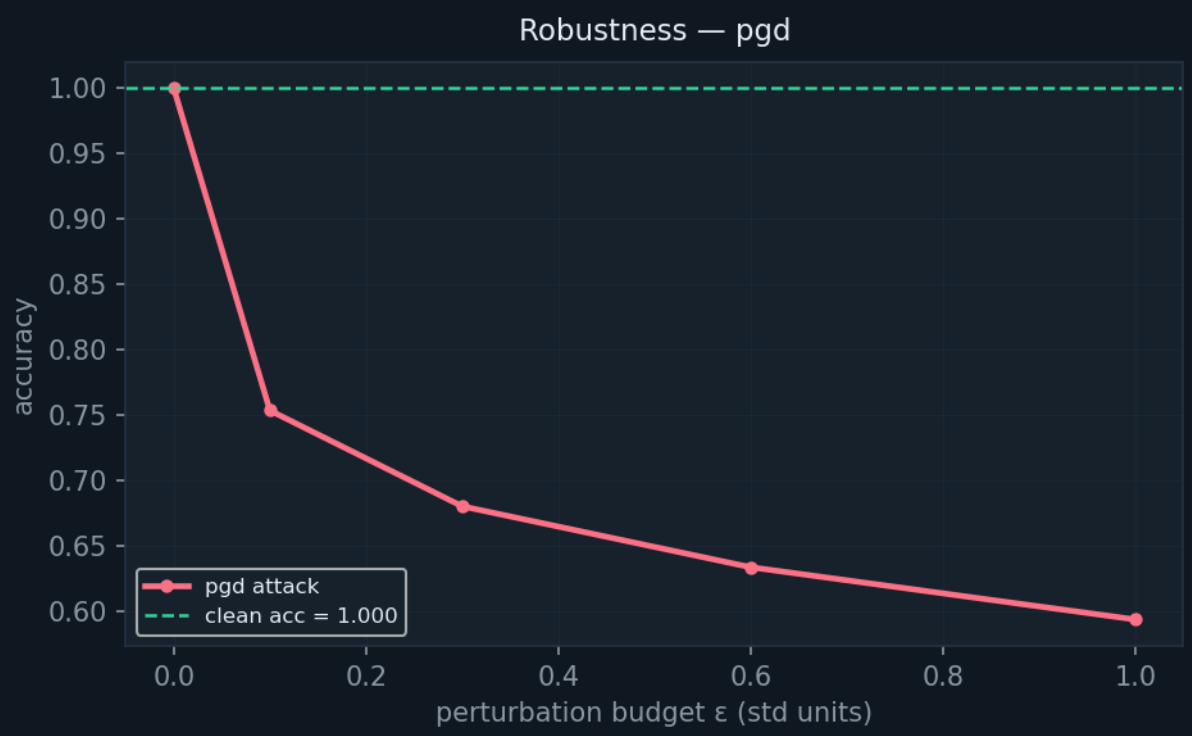
Metric	Value
Adv trained robust acc@eps0.6	0.9667



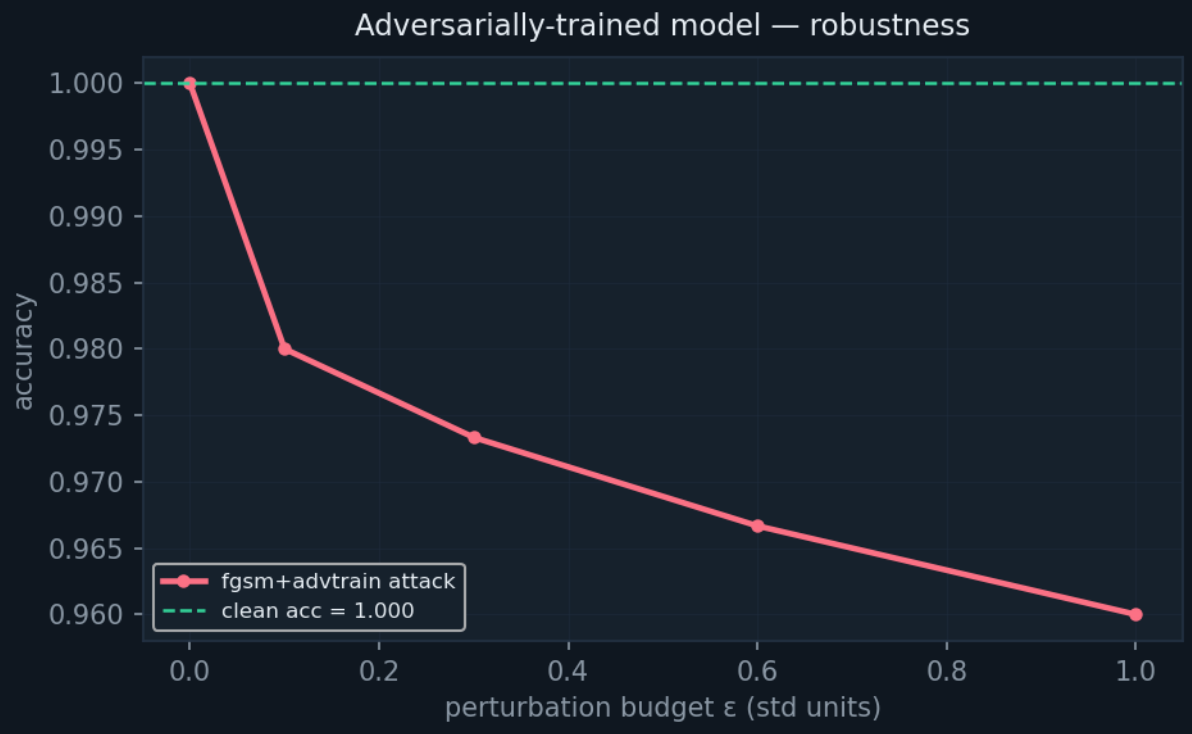
Attack comparison



Robustness fgsm



Robustness pgd



Adversarial training

Attacks

attack	eps	clean_accuracy	robust_accuracy	attack_success_rate	mean_perturbation
fgsm	0.3	1	0.4733	0.5267	0.3
pgd	0.3	1	0.68	0.32	0.3
deepfool	0	1	0.5333	0.4667	53.7

Robustness fgsm

epsilon	accuracy
0	1
0.1	0.4733
0.3	0.4733
0.6	0.4733
1	0.4733

Robustness pgd

epsilon	accuracy
0	1
0.1	0.7533
0.3	0.68
0.6	0.6333
1	0.5933

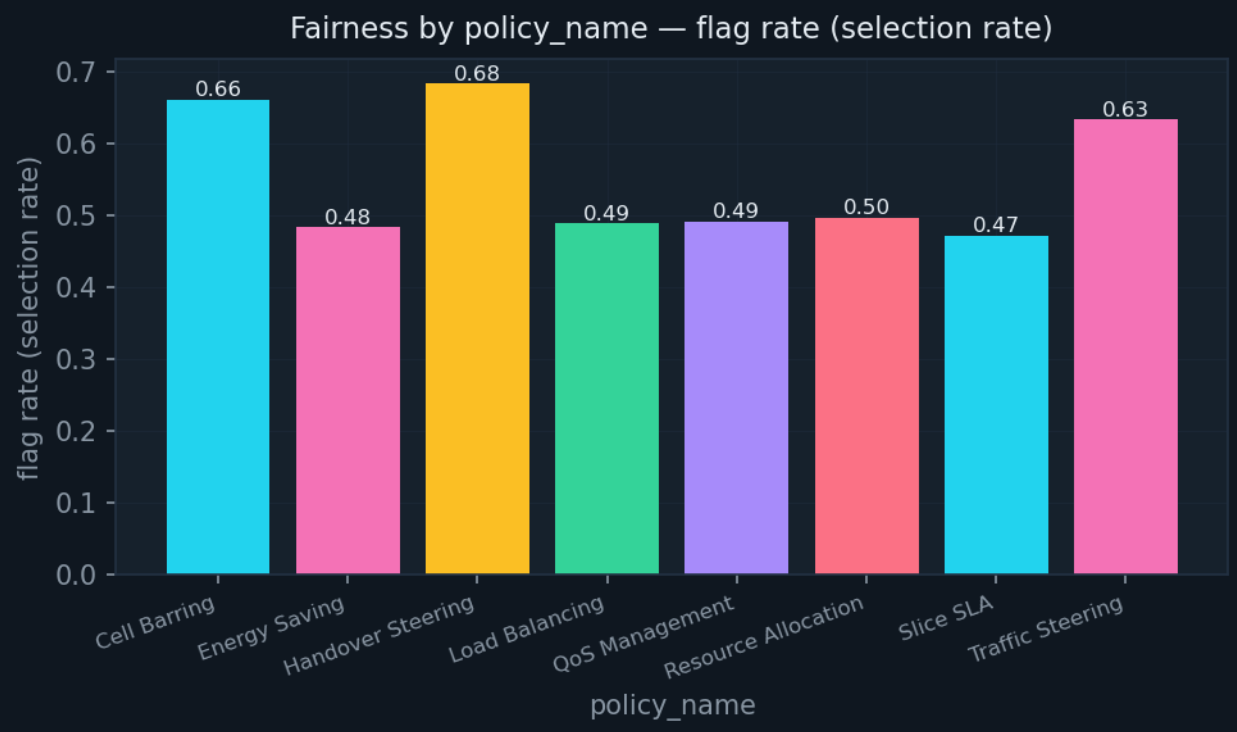
Adversarial training

epsilon	accuracy
0	1
0.1	0.98
0.3	0.9733
0.6	0.9667
1	0.96

Fairness

Selection and error rates per group, demographic parity, disparate impact and equalised-odds differences over 8 groups.

Metric	Value
Group attr	policy_name
N groups	8
Demographic parity difference	0.212
Demographic parity ratio	0.6897
Disparate impact	0.6897
Equal opportunity difference	0
Equalized odds difference	0.0116
Accuracy gap	0.0043



Fairness groups



Fairness metrics

Group metrics

group	n	selection_rate	tpr	fpr	fnr	accuracy	precision	positives
Cell Barring	186	0.6613	1	0	0	1	1	123

group	n	selection_rate	tpr	fpr	fnr	accuracy	precision	positives
Energy Saving	1330	0.4835	1	0.001453	0	0.9992	0.9984	642
Handover Steering	180	0.6833	1	0	0	1	1	123
Load Balancing	1312	0.4886	1	0.001488	0	0.9992	0.9984	640
QoS Management	1287	0.4918	1	0	0	1	1	633
Resource Allocation	1332	0.4962	1	0	0	1	1	661
Slice SLA	1341	0.4713	1	0	0	1	1	632
Traffic Steering	232	0.6336	1	0.01163	0	0.9957	0.9932	146

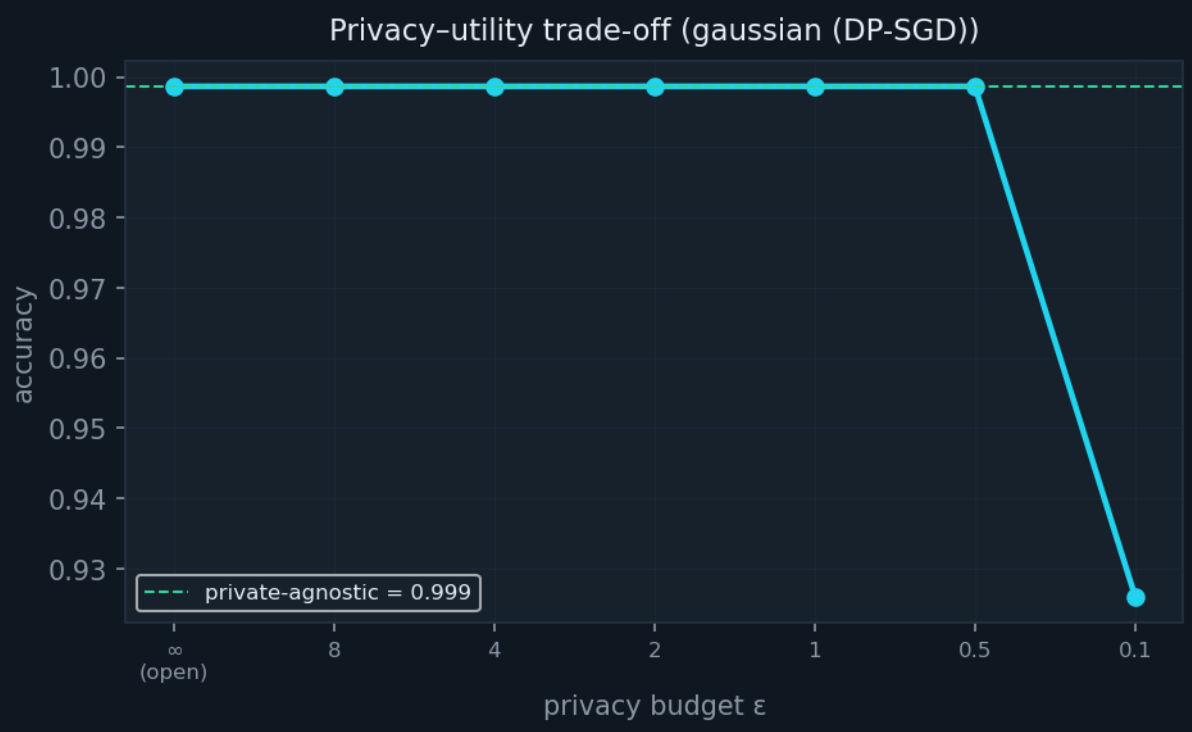
Disparity metrics

metric	value
Demographic parity difference	0.212
Demographic parity ratio	0.6897
Disparate impact	0.6897
Equal opportunity difference	0
Equalized odds difference	0.01163
Accuracy gap	0.00431

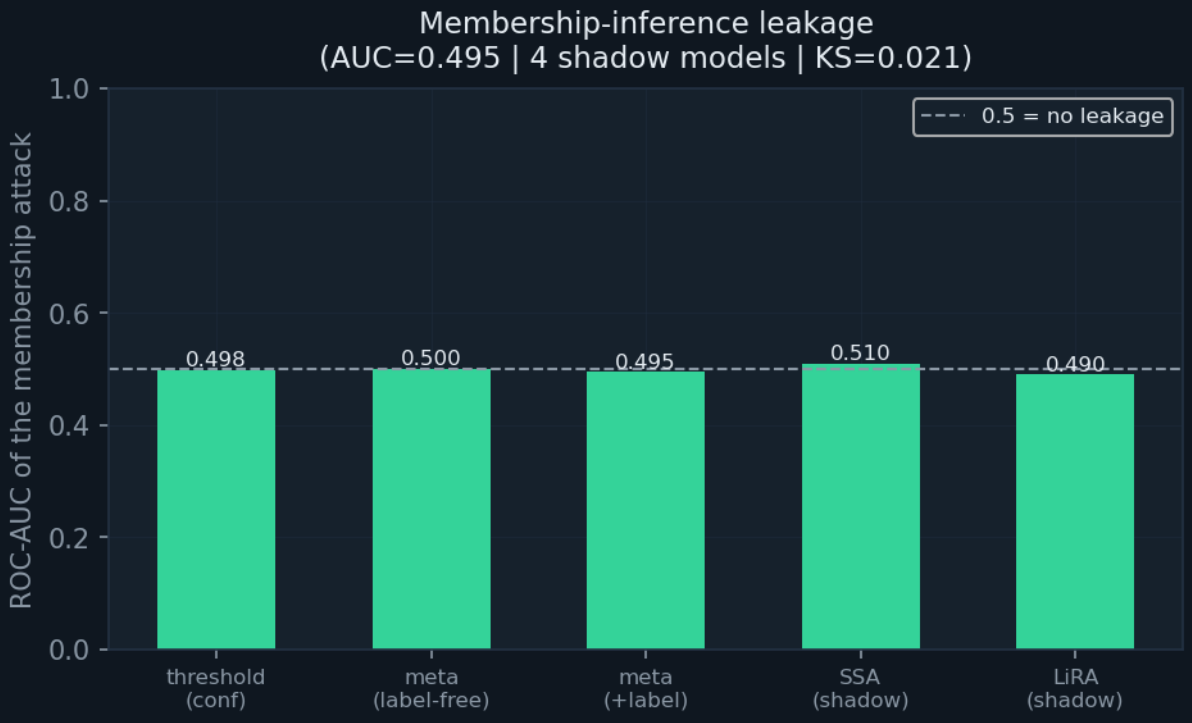
Privacy

DP-SGD with per-sample gradients and moments-accountant accounting over an ε sweep, FedAvg across — clients, and a membership-inference attack group (confidence threshold, label-aware meta-model, shadow SSA and LiRA).

Metric	Value
Dp mechanism	gaussian (DP-SGD)
Utility open	0.9987
Utility eps1	0.9987
Utility eps0.1	0.926
Membership inference auc	0.4946
Mia shadow auc	0.51
Mia lira auc	0.4904
Mia conf only auc	0.5
Mia ks	0.02133
Mia tpr at 1pct fpr	0
Mia n shadows	4



DP curve



Membership inference

DP curve

epsilon	utility
∞	0.9987
8	0.9987
4	0.9987
2	0.9987
1	0.9987

epsilon	utility
0.5	0.9987
0.1	0.926

Membership inference

metric	value
Attack auc	0.4946
Attack accuracy	0.5025
Auc threshold	0.4978
Auc lr conf	0.5
Auc lr meta	0.4946
Auc ssa	0.51
Auc lira	0.4904
Auc ssa accuracy	0.6662
Ks stat	0.02133
Tpr at 1pct fpr	0
Confidence gap	0.0001561
Generalization gap	0.0006667
Near duplicate rate	0
N shadows	4

O-RAN console snapshot

No console session state was captured.

Agent trust evaluation

No agent evaluation has been run yet — run it in the Agent Eval panel to fill this section.

Methodology & validity limits

Reproducibility

```
python -m trustoran --dataset cci_a1sd_binary --model xgboost --pillars all --quick
python -m trustagent --quick
```

Metric definitions

Clean accuracy	Accuracy before any perturbation is applied.
Fgsm robust acc	Accuracy under one-step FGSM at the reported ε.
Pgd robust acc	Accuracy after 10-step PGD optimisation in the ε-ball.
Deepfool robust acc	Accuracy under the minimum-perturbation DeepFool attack.
Pgd attack success rate	Share of rows whose true label is flipped by PGD.
Adv trained robust acc@eps0.6	Same model retrained on adversarial examples.
Demographic parity difference	Max minus min selection rate across groups.
Disparate impact	Min / max selection rate; the 4/5-rule quantity.
Equalized odds difference	Largest TPR/FPR gap across groups.

Clean accuracy	Accuracy before any perturbation is applied.
Accuracy gap	Best minus worst group accuracy.
Utility open	Utility of the unconstrained baseline network — accuracy for classification, R^2 for regression.
Utility eps1	Utility at a privacy budget of $\varepsilon=1$ — accuracy for classification, R^2 for regression.
Membership inference auc	Membership-attack AUC of the meta-model fitted on label-aware attack features; 0.5 means no leakage signal. The threshold, shadow (SSA) and likelihood-ratio (LiRA) attacks are reported next to it.

Validity limits

- Public research corpora — not production RAN telemetry.
- Attacks use differentiable surrogates, so robustness figures describe this attack family rather than an exhaustive adversary.
- Privacy accounting follows the moments accountant inside the local DP-SGD loop; no formal (ε,δ) certificate is issued for the shipped estimator.
- The fairness grouping is a modelling choice fixed per dataset in DATASET_PROFILES, not a property of the data itself.
- The agent environment is a sandbox with ACLs, budgets and confirmation gates; no real system or network element is touched.
- Mock-mode agent numbers validate the harness only; they are not a measurement of any real model.
- For normative wording, consult the O-RAN Alliance specifications directly.

This module is a personal research project that extends trustworthy-AI methodology to agent evaluation; it is not affiliated with any employer. The O-RAN scenarios and standards summaries are the author's own paraphrasing of public material.